Problem statement

Healthcare disparities are a persistent global challenge, impacting millions of lives through preventable deaths, inequitable access to care, and worsening economic and social inequities. While high-income countries benefit from advanced healthcare systems and infrastructure, middle and low-income countries struggle with resource shortages, limited access to services, poor health conditions, and significant financial barriers. These disparities stem from systemic challenges such as insufficient funding, workforce shortages, socio-economic inequalities, and the exacerbating effects of global crises like COVID-19 pandemic.

This project investigates the structural and systemic causes of healthcare inequities across 33 countries, representing high, middle, and low-income groups. Using 20 years of longitudinal data from the World Bank and 50 key indicators spanning healthcare resources, outcomes, expenditures, and public health metrics, this study aims to identify critical factors driving disparities and analyze how socio-economic and demographic variables influence health outcomes. The analysis will also examine the long-term effects of global crises, such as COVID-19 on healthcare systems. The objective is to uncover trends and root causes of inequities that can inform data-driven strategies for addressing healthcare disparities globally.

Articulation of Value

Addressing healthcare disparities is critical for improving health outcomes, reducing preventable mortality, and promoting global health equity. This project provides value by generating actionable insights that encourage policymakers, healthcare providers, and global organizations to allocate resources more effectively, prioritize interventions, and target underserved populations. The findings aim to support data-driven strategies to improve healthcare access and strengthen systems across countries with varying income levels.

The economic impact of addressing these disparities is substantial. Preventing avoidable hospitalizations and reducing the financial strain on individuals and governments can lead to healthier populations which contribute to greater workforce productivity and thus economic stability. Furthermore, building a robust healthcare system also equips countries to respond more effectively to future crises and minimize disruptions in care delivery during global emergencies.

This project contributes to long-term global health goals by supporting the development of equitable healthcare systems that foster better and reduce systemic barriers to care. The findings align with the United Nations Sustainable Development Goals (SDG 3: Good Health and Well-being) by advancing universal health coverage and addressing disparities that hinder progress toward achieving global health equity.

Calculation of the potential economic value

The calculations assume a standardized population of 10 million for comparability across income groups. However, in reality, country populations vary significantly. To ensure accurate estimates, the model will scale hospital admissions, preventable hospitalization rates, and economic value calculations proportionally based on each country's actual population size.

1. Healthcare cost savings: addressing preventable hospitalizations is a significant step to reduce healthcare spending. Countries can minimize unnecessary hospital admissions by optimizing resource allocation and improving access to preventive care.

Assumptions for hospital admission rates: hospital admission rates vary across income groups due to differences in healthcare infrastructure, disease burden, and access to inpatient care. Studies have shown significant disparities in hospitalization trends as:

- High-income countries: estimated 13% of the population is admitted to hospitals annually.
- Middle-income countries: estimated 10% of the population.
- Low-income countries: estimated 7% of the population.

Calculations: for a population of 10 million

- High-income: 10,000,000 × 0.13 = 1,300,000 admissions.
- Middle-income: 10,000,000 x 0.10 =x1,000,000 admissions.
- Low-income: 10,000,000 x 0.07 = 700,000 admissions.

Assumptions for preventable hospitalizations: the preventable hospitalization rates are general estimates. OECD and WHO reports indicates that actual preventable hospitalization rates can vary significantly depending on healthcare access, quality of primary care, and socioeconomic conditions. High-income countries have better preventive care with lower preventable hospitalization rates, while middle-income and low-income countries face greater barriers to primary care with higher rates.

- High-income: 5%.
- Middle-income: 10%.

- Low-income: 15%.

Calculations: for a population of 10 million

- High-income: 1,300,000 x 0.05 = 65,000 preventable admissions.
- Middle-income: 1,000,000 x 0.10 = 100,000 preventable admissions.
- Low-income: 700,000 x 0.15 = 105,000 preventable admissions.

Assumptions for average cost per admission: this assumptoon varies significantly based on healthcare infrastructure, service quality, and economic conditions. These estimates are derived from reported healthcare expenditures and hospital costs across different regions and serve as a general benchmark for comparison across income levels."

- High-income: $5,000 per admission.
- Middle-income: $1,000 per admission.
- Low-income: $500 per admission.

Calculations: for a population of 10 million

- High-Income: 65,000 x 5,000 = $325,000,000.
- Middle-Income: 100,000 x 1,000 = $100,000,000.
- Low-Income: 105,000 x 500 = $52,500,000.

In terms of economic value, high-income countries may contribute $325 million annually, middle-income countries can contribute $100 million annually and low-income countries contribute $52.5 million annually.

2. Workforce productivity gains: improved health outcomes reduce illness-related absenteeism, thus increase workforce productivity and economic contributions.

Assumptions: 1% of the population benefits annually from improved healthcare access.

- GDP contribution per worker for high-income: $70,000 annually.
- GDP contribution per worker for middle-income: $20,000 annually.
- GDP contribution per worker for low-income: $5,000 annually.

Calculations: For a population of 10 million:

- Population with health improvements: 10,000,000 x 0.01 = 100,000.
- Productivity Gains for high-income: 100,000 x 70,000 = $7,000,000,000.
- Productivity Gains for middle-income: 100,000 x 20,000 = $2,000,000,000.
- Productivity Gains for low-income: 100,000 x 5,000 = $500,000,000.

In terms of economic value, high-income countries may contribute $7 billion annually, middle-income countries can contribute $2 billion annually, and low-income countries contribute $500 million annually.

3. Health expenditure growth: healthcare spending growth shows the increasing costs for countries and underscores the need for efficient spending and focused actions to manage expenses and improve outcomes

Assumption: based on WHO and World Bank reports, healthcare expenditures grow annually by 2% in high-income countries, 3% in middle-income countries and 5% in low-income countries.

Calculation: $expenditure = initial\ expenditure * \ (1 + growth\ rate)^{years}$

For a country with initial expenditure of $50 billion and annual growth

- High-income: $50{,}000{,}000{,}000 * \ (1 + 0.02)^{10}$= 60,950,000,000
- Middle-income: $50{,}000{,}000{,}000 * \ (1 + 0.03)^{10}$= 67,150,000,000
- Low-income: $50{,}000{,}000{,}000 * \ (1 + 0.05)^{10}$= 81,450,000,000

In terms of economic value, the new expenditure after 10 years is $60.95 billion for high-income countries, $67.15 billion for middle-income countries, and $81.45 billion for low-income countries.

4. Impact of healthcare interventions: effective healthcare interventions can reduce preventable hospitalizations, leading to improved economic outcomes.

Assumption: WHO studies show that primary care and preventive measures can reduce avoidable hospitalizations by 2%.

Calculation: $\text{Preventable hospitalizations} = \text{initial preventable admissions} * \ (1 - \text{intervention impact rate})^{years}$

For a country with initial preventable admissions of 100,000 over 10 years:

- $100{,}000 * \ (1 - 0.02)^{10}$= 81,700 for all high, middle, and low countries

Cost savings = (initial admissions - adjusted admissions) * cost per admission.

- High-income: (100,000−81,700) * 5,000 = $91,500,000
- Middle-income: (100,000−81,700) * 1,100 = $20,130,000
- Low-income: (100,000−81,700) * 109 = $1,994,700

In terms of economic value, healthcare interventions result in annual cost savings of $91.5 million in high-income countries, $20.13 million in middle-income countries, and $1.99 million in low-income countries over 10 years.

5. Productivity loss from illness: provides a more comprehensive view of how poor health affects economic growth.

Assumption: studies from WHO and the ILO indicate that health-related absenteeism and presenteeism contribute significantly to lost productivity about 15% annually.
Calculation: lost productivity = population affected * average gdp per worker * productivity loss percentage. For a country with a population affected about 10% of the total workforce and productivity loss of 15%.

- High-income: 1,000,000 * 70,000 * 0.15 = 10,500,000,000
- Middle-income: 1,000,000 * 20,000 * 0.15 = 3,000,000,000
- Low-income: 1,000,000 * 5,000 * 0.15 = 750,000,000

In terms of economic value, the total loss amounts to $10.5 billion for high-income countries, $3 billion for middle-income countries, and $750 million for low-income countries annually.

Footnotes

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WHO & World Bank. Global Trends in Health Expenditures.
WHO. Global Workforce and Productivity Studies.
Health Economics Review. Costing Health Services in Jordan.
WHO. Impact of Healthcare Interventions on Population Health.
WHO. Impact of Healthcare Interventions on Hospitalizations.
PMC. Costs of ICU Care in India.

Project plan

Week 1: finalize the problem statement and articulate the significance.

- Define the scope and objectives of the project.
- Select the dataset aligned with the problem statement.
- Finalize the project plan and timeline.
- Set up a GitHub repository.

Week 2: understand the dataset and prepare it for analysis.

- Perform data cleaning
- Add derived variables, for example regional groupings, external factors like COVID-19.
- Document findings.

Week 3: data preparation and exploratory analysis for understanding healthcare disparities

- Divide the dataset into training, validation, and test subsets for robust analysis.
- Conduct exploratory data analysis (EDA) to uncover patterns and key insights.
- Identify and address data challenges like missing values, outliers, and imbalances.

Week 4: data cleaning and preparation for model readiness

- Clean the healthcare dataset by addressing missing values, outliers, and inconsistencies.
- Process the data, including encoding categorical variables and normalizing features.
- Prepare the dataset for analysis, ensuring it is model-ready.

Week 5: feature engineering and dimensionality reduction

- Engineer new features from healthcare indicators to uncover additional insights.
- Assess the need for data augmentation and apply techniques if required to balance the dataset.
- Reduce dimensionality using methods like PCA to focus on key factors driving healthcare disparities.

Week 6: model training and optimization

- Train initial models to analyze healthcare disparities using the processed dataset.
- Evaluate training and validation errors to detect overfitting or underfitting.
- Optimize hyperparameters while prioritizing simple, interpretable models.

Week 7: model refinement and complexity trade-offs

- Refine models by analyzing training/validation errors and adjusting parameters.
- Explore the trade-offs of using more complex models for understanding healthcare disparities.
- Balance interpretability and performance to ensure practical insights for policy and intervention.

Week 8: model evaluation and selection

- Build models with varying complexity and test different hyperparameter settings.
- Define success metrics tailored to healthcare disparity analysis, such as fairness or accessibility measures.
- Evaluate model performance using the validation dataset to select the most effective approach.

Week 9: final model evaluation and testing

- Evaluate models on the validation dataset to assess their ability to explain healthcare disparities.
- Select the best model based on accuracy, interpretability, and relevance to global healthcare systems.
- Test the selected model on unseen data to predict performance and ensure generalization.

Week 10: data-centric AI and dataset improvement

- Explore data-centric AI to address healthcare disparities by emphasizing data quality.
- Refine the dataset to ensure diverse and accurate representation of global healthcare systems.

Week 11: explaining predictions and analyzing bias

- Explain model predictions and identify the most important features driving outcomes.
- Quantify and address biases in the dataset and model outputs.
- Analyze model risks to ensure reliability and fairness in healthcare disparity analysis.

Week 12 Plan: deployment and monitoring

- Package the selected model for deployment in healthcare applications.
- Develop a monitoring plan to track the model's performance in real-world use.
- Build a plan to address data and concept drift, ensuring long-term model reliability.

Week 13 plan: end-to-end project completion

- Create complete source code for the end-to-end modeling pipeline.
- Develop a narrative that summarizes the entire project, highlighting key findings and methods.
- Organize the GitHub repository for effective sharing and collaboration.

Dataset

The dataset for this project is drawn from the World Bank Health, Nutrition, and Population Statistics. We selected indicators to address the challenge of global healthcare disparities. The dataset includes 51 indicators, grouped into six categories: healthcare resources, health outcomes, healthcare expenditures, socio-demographic factors, public health metrics, and water, sanitation, and hygiene (WASH).

The selected data spans 20 years, from 2002 to 2021, we aim to provide a longitudinal view of trends in healthcare systems. We will analyze how disparities have evolved over time and examine the impact of external events, such as the COVID-19 pandemic on healthcare delivery. The dataset includes 33 countries, categorized into high, middle, and low-income groups to ensure that our analysis captures diverse healthcare systems and highlights the inequalities that exist across income levels.

We believe that this dataset is highly relevant to solving our problem statement, given its global scope and longitudinal structure. By examining the relationships between different categories, we can identify the key factors driving healthcare disparities. With this data, we aim to generate actionable insights that can inform policies and interventions to reduce inequities. Ultimately, our goal is to support equitable resource allocation, strengthen healthcare systems, and improve health outcomes for diverse populations worldwide.

Models

This project will both supervised learning and time series modeling to address the problem statement. For supervised learning models, we aim to analyze the relationships between healthcare resources, socio-economic factors, and outcomes. We will start with linear regression for simple relationships. Next, we will use Random Forest Regression to capture non-linear interactions and understand feature importance. Gradient Boosting Machines will be applied to handle more complex relationships and improve predictive accuracy. Finally, Lasso Regression will help identify the most influential indicators by reducing less important variables.

For time series models, we will use ARIMA to forecast single variables like life expectancy or healthcare expenditures. ETS will model trends and seasonal variations in key indicators. Additionally, multivariate time series models will be used to study interdependencies between variables such as expenditures, resources, and outcomes.